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Multi-scale modelling of monolith reactors: A 30-year perspective from 1990 to 2020

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Abstract

The honeycomb type monolith reactor dominates the field of catalytic automotive exhaust gas after treatment systems and is increasingly being used in other applications. Advanced computational modelling of such systems can provide valuable insight into their operating characteristics and be used as a design tool. This reactor provides an excellent example of a multi-scale modelling challenge with many complexities. This paper provides a perspective on the development and use of computational modelling tools in this context over the last 30 years, with an emphasis on the work published by the researchers at the University of Alberta. This paper covers all aspects of model development, including single channel and full-scale reactor models and inter and intra mass and heat transfer effects, with the latter including radiation.

KEYWORDS

catalytic converter, CFD, modelling, monolith

1 | INTRODUCTORY EXPLANATORY NOTE BY R. E. HAYES

When I was asked by Professor João B. P. Soares to submit a paper to the special series highlighting the work of Canadian researchers, it seemed natural to write about developments in the modelling of structured honeycomb type monolith reactors. I started work on modelling this reactor in the summer of 1990, early in my career, having been invited by the late Professor W. J. Thomas of the University of Bath to participate in a collaborative effort on the use of computational modelling to enhance the understanding of the behaviour of monolith reactors. This invitation arose from conversations over a pub lunch at the Wheelwrights Arms in Monkton Combe some two years earlier, during a flying visit to the University of Bath on my way to the 1988 International Symposia on Chemical Reaction Engineering in Switzerland. This collaboration with the University of Bath, first with Professor Thomas and later with Stan Koloczkowski, initially focused on catalytic combustion for turbine applications, and afterwards, through involvement with Degussa, later DMC[2], then OMG, and finally Umicore, in the area of automotive exhaust gas after treatment systems (EGATS). Most of my work contained a major modelling component, with an emphasis on heat and mass transfer in its various guises. It is interesting to look back over the last 30 years (which seems like a short time to me) and note the significant advances that have been made in the area of monolith reactor modelling. In the year 2000, I made a keynote presentation at the 2nd Canada-Korea Workshop on Environmentally Friendly Technology in Montreal, entitled 'Catalytic Combustion Modelling: A 10 Year Perspective'. That work summarized the advances over the previous 10 years. It therefore seemed natural, two decades later, to make this paper a 30-year perspective.

This paper provides a historical perspective and review of this topic for that period, focusing on momentum, mass, and heat transfer effects, in the presence of chemical reactions. The review is not meant to be comprehensive, and there are many other studies that have contributed to the advancement of this field. However, adhering to the concept of this special series, the focal point of the article is the relevant papers published by the Hayes group at the University of Alberta, in collaboration with others, to emphasize how these papers have made significant improvements in our understanding. This paper is divided into the important topics, each of which we have tried to present in a chronological order, in so far as is possible. Work from the group also includes much effort on kinetic modelling, but that topic will only be referred to in the context of applications of the monolith models.

2 | BACKGROUND

2.1 | Reaction engineering and the monolith reactor

To wax philosophical, we observe that chemical reaction engineering, especially catalysis, makes life on Earth at the present level possible. For example, the Haber process for fixing nitrogen allows the planet to feed its roughly eight billion inhabitants. At a smaller scale, the development of fluidized catalytic cracking before and during the Second World War made a significant contribution to an Allied victory and led the post-war development and expansion of reliable transportation via the automobile industry. Following the law of conservation of misery, the rapid rise in the use of automobiles led to increases in pollution, with significant reductions in air quality. The health risks from this pollution led to increasingly stringent emission regulations; first for carbon monoxide and hydrocarbons, followed by oxides of nitrogen, and then particulate matter. Many mitigation solutions were tried in the early days, before the universal adoption of catalytic EGATS during the 1970s. The catalytic systems that have resulted are an amazing feat of chemical reaction engineering, surpassing in complexity even the Haber process, especially considering the three-way-catalyst (TWC) for stoichiometric gasoline engines. The underlying catalytic systems for various engines are not the topic of this paper; the reader is referred to the book by Heck et al.[1] for an excellent overview and history.

In the early days of catalytic EGATS, several types of reactor were used, before the washcoated honeycomb type monolith design was universally adopted. The monolith honeycomb consists of a metal or ceramic substrate that contains thousands of parallel channels, typically with a hydraulic diameter of the order of 1 mm. The catalyst is contained in a thin layer of washcoat that is applied to the walls. The monolith structure offers the advantages of low backpressure with no flow bypassing, good structural integrity, and a high geometric surface area per unit volume. The straight through channel design allows for the easy passage of entrained particulate matter which, unless it is desired to capture this matter, prevents fouling of the reactor.[2]

The inherent advantages of monolith reactors over, for example, traditional packed beds, have led to other applications. There was interest from the late 1980s through the mid-1990s in using catalytic combustion for thermal energy generation, especially for turbines, and monoliths were the reactor of choice.[3] Catalytic combustion has been used for fugitive methane emission destruction.[4] Since then, they have been applied to such processes as Fischer-Tropsch synthesis,[5,6] steam reforming,[7] partial oxidation of CO (CO-PROX),[8] and catalytic combustion of volatile organic compounds (VOC).[9]

2.2 | Computational tools

One of the most interesting observations in the story of EGATS is that their early development was almost entirely experimentally based. Much significant work in EGATS was done in the late 1960s and into the 1970s. Reflecting back on the computational power of those days, comparing both hardware and software to those in common use today, and the concomitant development of the underlying mathematical methods, the advances have been monumental. Whilst it would be an exaggeration to say that the technology then was not much beyond the level of 'stone knives and bearskins',[10] the ability to solve complex systems of partial differential equations (PDE) was somewhat limited. In 1990, the personal computer was not a useful tool for any sort of advanced computation, although it had improved from its proverbial early role as a place to store recipes. While the use of advanced numerical methods in the form of computational fluid dynamics (CFD) was gaining ground, it was not widely used in the chemical process industries, including most aspects of reactor analysis, and, as noted, computational complexity was a major constraint on its widespread use. At the same time, commercial CFD type software was limited in scope and availability. Reflecting over the last 30 years, there have been huge advances in all computational areas, which is reflected in the improvements in our models.

In light of these factors, our modelling work in the context of monolith reactors has changed with the times, and mirrors the methodology followed by other research groups. From 1990 to about 2003, simulations were done using our own software written in C and Fortran. These programs used the finite element method (FEM) to solve the governing PDE. In 2003 we started experimenting with FEMLAB (later COMSOL MultiPhysics) and shortly thereafter with Fluent. After 2005 we were using our own software, mostly for kinetic modelling studies only (along with MATLAB, DVODE, and the NAGS libraries), and after 2010 most of our research and resulting publications have used commercial simulation software.

2.3 | Modelling the monolith: The multi-scale problem

To appreciate the full extent of model development over the last 30 years, it is necessary to have a minimal appreciation of the types of models that can be used for monolith reactors, and the multi-scale nature of the problem. As noted earlier, the monolith substrate for EGATS applications consists of parallel channels about 1 mm in hydraulic diameter. The catalyst is contained within a washcoat that covers the channel walls. A catalytic converter for even a small displacement engine has thousands of channels, with a diameter of the order of tens of centimetres. Reactors for other processes will similarly contain many thousands of channels. Realizing that the catalytic washcoat has pores starting in the 10 nm range, it is seen that the multi-scale nature of the problem ranges over at least seven orders of magnitude. Each spatial scale typically requires the solution of a set of governing PDEs for the conservation equations. If we consider the spatial scales, a model of high granularity would require the complete domain discretization down to the smallest scale, which obviously would give a very large problem requiring vast computational resources, not to mention challenges in mesh generation. Therefore, model reduction strategies are used, with the various scales being incorporated using sub-models (also called scale bridges). Much of the advancement of monolith reactor models is related to the development of these sub-models, and how they are incorporated into the master model. Keeping in mind that all models contain error in some form, whilst at the same time they should be made as simple as possible, we should select our models and scale-bridges with due diligence.

In general, monolith models can be divided into two main types. The first is the so-called single channel model, or SCM, and the second is a model for the entire reactor system, here called ECM. In the following sections we introduce the SCM and the ECM and discuss the issues and challenges in each. We illustrate how the models have evolved to the current state of the art.

3 | THE SINGLE CHANNEL MODEL: SCM

3.1 | Overview and challenges

The SCM represents a single channel of the monolith. Considering only a single channel allows for much greater spatial detail to be included in the model, although SCMs come in varying levels of complexity. SCMs can be particularly useful in revealing some of the fine scale details, and as an aid in the development of scale-bridges.

Ceramic monoliths come in a variety of cross-sectional shapes, with square being the most common, followed by hexagonal. The catalytic washcoat usually varies in thickness around the perimeter of the channel, with a typical range from 10–150 μm in thickness, depending on the channel shape and coating method. A perspective view of a single channel of a monolith with square channels and a non-uniform washcoat is shown in Figure 1.

With a washcoat layer of non-uniform thickness, the real structure of the channel requires that a full spatially 3D model be used, although with symmetry we can consider one-eighth or one-quarter of the channel, provided that laminar flow is assumed. Fully 3D models are rare in the literature, largely because of the high computational cost involved in their solution. If one performs a geometrical reconfiguration (see Figure 2), then the 3D structure can be represented as a right circular cylinder, and a 2D axi-symmetric model can be used instead. This model requires a washcoat of uniform thickness, and thus an average value would be used. Probably the best way to make this transformation is to preserve the cross-sectional areas of the open channel, substrate structure, and washcoat. Some error is introduced in this simplification.

2D and 3D models use the same conservation equations for momentum, mass, and energy. They are

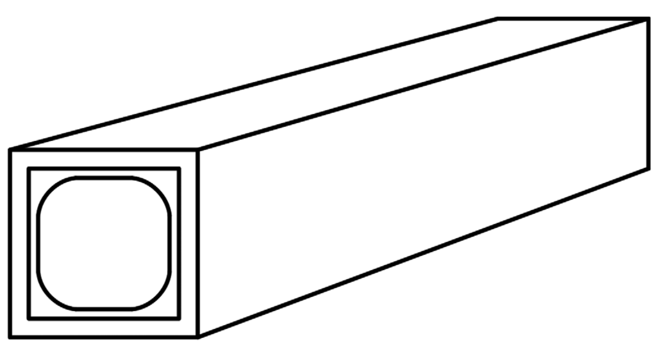

FIGURE 1 Illustration of a washcoated square monolith channel

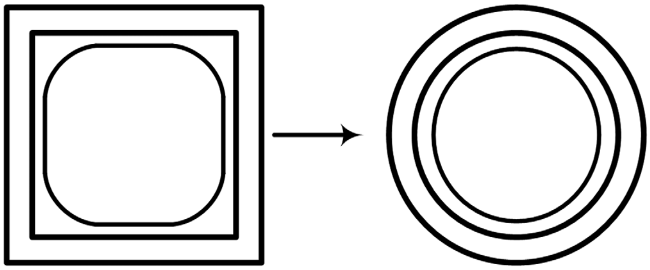

FIGURE 2 Geometrical transformation from square to circular domain

distributed parameter models with axial and radial temperature and concentration gradients. The solution domain includes the fluid-filled channel, the washcoat, and the wall. Model reduction strategies might be used to reduce the complexity of the domain, as discussed shortly, although the washcoat and wall must be accounted for either explicitly or implicitly.

Further simplification of the spatial dimensions in the model requires a change from a distributed parameter to a lumped parameter model. We can ignore radial gradients in both temperature and concentration and use their mixing cup averages, giving a 1D model for the fluid. A flat velocity profile is assumed. At the washcoat interface the discontinuity is accounted for using heat and mass transfer coefficients. To simplify further, one could even move to a completely lumped homogeneous model, where all heat and mass transfer resistances are ignored and everything is evaluated at the bulk mean temperatures and concentrations. It is important that the washcoat and substrate be accounted for in a consistent fashion.

Each of the assumptions and resultant choices require decisions, which to a large extent depend on the desired objectives of the model combined with the available computational resources. The choices have an impact on the form of the momentum, energy, and mass balance equations, both in the fluid phase and the porous catalytic washcoat. We discuss model development in the context of each of these balances.

We now say a brief word about the state of SCM in 1990, as the start of the period. It is not our intention to provide a comprehensive literature review, but simply to give a feel for the state of the art. Although arguably not the first paper on modelling of monolith channels, two publications in 1976 illustrate the methodology used at that time.[11,12] These papers were fairly comprehensive for their time and can be consulted for a discussion of the limited earlier work. Both 1D and 2D models were compared, although some assumptions were made in each case, which are further discussed later in the paper. Unfortunately, they do contain some generalizations, presented as truisms, which may have contributed to later misinterpretation.

3.2 | Modelling the velocity distribution

The momentum balance equation gives information about velocity and pressure distributions in the monolith channel. Clearly, a 1D model will not give any information here; indeed, a flow model must be assumed if a pressure drop is to be determined. The common assumption in 1D models is to assume fully developed laminar flow, and the pressure drop can be calculated from the Hagen-Poiseuille equation. If we move to a 2D or a 3D domain, then there are two possibilities. Before 1990, models in 2D[11–14] assumed fully developed laminar flow over the entire channel and imposed a velocity profile. The Navier–Stokes equation was not solved. This concept is equivalent to solving the Graetz problem (discussed later), and the drawback is that the entry length is not treated correctly. This might not be significant for a long converter, and our own 2D model in its first incarnation made this assumption.[15,16] The radial and axial velocity profiles were corrected for the local temperature using the ideal gas law. Almost immediately, the model was extended to include the solution to the momentum balance equation for a variable density fluid.[17–21] We should also note that from the beginning, our models used temperature dependent physical properties, which later was seen to be quite significant, because it allowed insight into the external heat and mass transfer, discussed shortly. Most 2D and 3D models published today solve the momentum balance.

In common with others, we imposed a flat velocity profile as the inlet boundary condition. In fact, it can be important to include the upstream domain if it is desired to capture the entrance length correctly. More[22] performed a detailed study of the SCM and included an upstream reservoir. The hydrodynamic entry length was shown to be different from the case of flat inlet velocity, especially for low Reynolds number. The developing flow solution is bounded by two asymptotic solutions for low and high Reynolds regions, respectively. At high Re the entrance length depends linearly on Re, whilst at low Re it is constant. The two solutions were combined using the asymptote matching technique[23] to give a solution:

$$\frac{L_{\mathrm{ent}}}{D} = \left[(0.54)^{1.62} + \left(0.05Re(\;)^{1.62}\right)[\;]^{\frac{1}{1.62}}\right] \tag{1}$$

A similar result was also published later.[24]

The extended straight reservoir does not account for all of the inlet effects in a monolith because the flow into

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the channel faces a reduction of surface area, thus the inlet should by modelled as flow into a contraction. We have shown that the inlet velocity profile is quite different for the three cases of no reservoir, straight reservoir, and a contraction.[25] These effects must be modelled correctly to capture the entry length heat and mass transfer effects.

3.3 | Modelling radiation

Radiation heat transfer along the axial direction of the monolith can increase the axial heat transfer rate, leading to a reduction in the temperature gradients. Most of the literature on monolith reactor modelling does not include radiation. In the early days, including radiation was expensive and quite complicated, although there are early examples of it, with two very good papers published in the late seventies.[14,26] Influenced by these two works, our first model[15] included radiation. Rather than adopt the mathematical approach in the earlier two papers, the radiation used a network analysis of finite grey isothermal surfaces with corresponding view factors,[27] with each surface within the monolith channel being comprised of the side of one finite element. This method requires only the solution of a matrix to determine the wall radiation fluxes. As noted, the main effect of radiation is to increase the effective axial thermal conductivity. There is, however, a large effect at the inlet and exit region, depending on the surrounding temperature. The magnitude of this effect was illustrated in a later paper,[18] in which our model predictions were compared to experimental axial temperature profiles, and we showed that radiation had to be included to capture the axial profile correctly.

In experimental measurements of temperature in a monolith, it is common to insert a thermocouple into a channel. The difference between indicated and actual temperatures for thermocouple measurements is always an issue, and we have shown that for a thermocouple that does not block the channel, the indicated temperature can be significantly affected by radiation.[19] Overall, radiation effects tend to be more significant at higher temperatures, such as found in catalytic combustors for energy generation, rather than lower temperature EGATS reactors. Finally, radiation can play a significant role in the static cooling of monoliths, that is, under conditions of no flow.[28] A summary of radiation techniques can be found in our book.[3]

We should note that current commercial modelling software, such as COMSOL and Fluent, incorporate complex radiation models, so accounting for it is more straightforward.

3.4 | Modelling the external heat and mass transfer

As noted in the preceding section, 2D and 3D models naturally account for the transfer of the energy and species to the washcoat surface. The use of an approximate 1D model, on the other hand, requires appropriate values for the heat and mass transfer coefficients between the bulk gas and the wall. The first major use for the 2D SCM developed in our group was to use it as a diagnostic tool to settle some controversy surrounding the correct selection of heat and mass transfer coefficients for use in a 1D model. Gas phase monolith reactor operation covers a wide range of possible Reynolds numbers. For EGATS applications, as well as many others, the Reynolds numbers are low, corresponding to laminar flow. In laminar flow, the monolith channel can be divided into the entry length, where the momentum, temperature, and concentration boundary layers are developing, and the fully developed region. The entry length might be significant, and will depend on the upstream flow, which is often turbulent. Furthermore, the flow may enter the channels at an angle. Finally, it is important to make the distinction between a local value and the average value of the heat or mass transfer coefficient; in our case, it is the former that is required for a correct model of the monolith channel. Therefore, it is fair to say that the problem is thus to determine local values for the Nu and Sh under conditions of developing and fully developed laminar flow in a channel of non-uniform cross-sectional shape with a catalytic reaction occurring at the wall. We now review how the understanding of this phenomenon has evolved over the last 30 years.

Heat transfer in laminar duct flow is covered in detail in three specialized books[29–31] and to some extent in basic heat transfer textbooks.[27] A good review of literature relevant to monolith reactors up to about 2005 is given in More.[22] In 1990, one school of thought apropos Nu in monolith channels under laminar flow was that the solution should not deviate significantly from the so-called Graetz problem. A competing view, which was based on experimental evidence, caused a certain amount of confusion in the literature.

The classical Graetz problem applies to an incompressible fluid flowing in a circular duct with fully developed laminar flow, a developing temperature profile, constant physical properties, and a constant wall temperature.[29] Some classical correlations based on Graetz theory are available.[27] For laminar fully developed flow in circular ducts, the Nu equals 3.656 for constant wall temperature and 4.364 for constant wall flux, with other values for non-circular shapes.[29] These values do not depend on Reynolds number.

In the 1990–1991 timeframe, the most commonly used Nu correlation of the Graetz type was the one proposed by Hawthorn,[32] based on work at constant wall temperature.[30] The correlation gave average values for Nu, not local ones. The equations for Nu and Sh are as follows:

$$\mathrm{Nu} = 3.66\left\{1 + 0.095\frac{D}{L}Re\ Pr\ \{\ \}^{0.45}\right\} \tag{2}$$

$$\mathrm{Sh} = 3.66\left\{1 + 0.095\frac{D}{L}Re\ \mathrm{Sc}\right\}^{0.45} \tag{3}$$

The competing line of thought was based on data from chemical vaporization experiments reported by Votruba et al.[33] The proposed correlations were as follows:

$$\mathrm{Nu} = 0.571\left(Re\frac{D}{L}\right)^{2/3} \tag{4}$$

$$\mathrm{Sh} = 0.705\left(Re\frac{D}{L}\right)^{0.43}\mathrm{Sc}^{0.56} \tag{5}$$

This summary describes the situation faced when we were developing our first 2D model, and there was much discussion within our group as to which model was correct. As a researcher and teacher in the area of heat transfer, it did seem rather obvious that the Graetz camp had a more theoretical underpinning than the Votruba camp. Although the Hawthorn correlation was clearly not appropriate for 1D SCM modelling, primarily because it was limited to constant wall temperature and yielded average values for Nu, it at least gave a reasonable asymptotic limit. In our first paper published on monolith reactors,[15] our new 2D model gave asymptotic values for Nu that were of the same order as those predicted from Graetz theory, which were also reasonably consistent with other theoretical work.[12,14,34,35]

On the other hand, the Votruba model gave low values for Nu and Sh, did not predict an asymptotic value, and the Nu and Sh depended on the Reynolds number, which does not reflect reality for laminar flow. In spite of this, there were other reported experiments that gave results consistent with the Votruba model, at least in terms of the order of magnitude. In particular, two investigations from the Bath group required explanation. In the first, Bennett et al.[16,36] studied oxidation of propane in a monolith and obtained the following correlations:

$$\mathrm{Nu} = 0.0767\left\{1 + \frac{D}{L}RePr\right\}^{0.829} \tag{6}$$

$$\mathrm{Sh} = 0.0767\left\{1 + \frac{D}{L}Re\ \mathrm{Sc}\right\}^{0.829} \tag{7}$$

Obviously, these equations predict low asymptotic values compared to the Graetz solution. Ullah et al.[37] conducted experiments under reacting conditions and reported values for Sh similar to Votruba:

$$\mathrm{Sh} = 0.766\left(\frac{D}{L}Re\ \mathrm{Sc}\right)^{0.483} \tag{8}$$

The assumptions and analysis technique employed were the same in both studies.

Investigating this discrepancy seemed like a good idea at the time, which led to the publication of a significant paper on monolith modelling.[17] Clearly, we had an initial bias, because all of the fundamental theory favoured a Graetz type solution, so we employed the scientific method to demonstrate that the experiments of Bennett et al.[36] and Ullah et al.[37] were not being interpreted correctly. Before continuing, it is essential to understand another point of received wisdom from that era, especially for highly exothermic oxidation reaction reactions in monoliths. At low temperatures, the reaction is slow, and is said to be kinetically controlled. As the temperature increases, there may come a point where the reaction is seen to ignite, that is, undergo a very rapid rise in reaction rate and concomitant release of thermal energy. Depending on the relative rates of mass transfer and reaction, the external mass transfer could become the controlling step, and the surface concentration would drop to zero. It is well known that some industrial processes, for example, the oxidation of ammonia on gauze catalysts, work this way. It was widely believed that catalytic combustion in monolith reactors also operated this way,[11] giving rise to the classical graph illustrating the operating regimes of such reactors (see, for example, figure 2 in Prasad et al.[38]). Whilst it can easily be shown by modelling that monolith reactors can work this way, it was accepted that they do work this way. Indeed, this assumed transition was associated with the so-called light-off point, which was defined as the inlet reactor temperature where the outlet conversion was 50%.[39] Although conventional wisdom at the time, there is no reason to associate an arbitrary conversion with a sudden change in reaction controlling regime, and in any case the light-off point so defined will depend on any number of operating conditions, including gas hourly space velocity (GHSV) and inlet temperature ramp rate.[3]

At this point, we needed to illustrate an alternate yet plausible set of conditions that could generate the same result as observed experimentally. The key assumption in both the Bennett et al.[36] and Ullah et al.[37] analysis was that the surface concentration of the reactant was essentially equal to zero, that is, complete external mass transfer control. Our first step was to use the full 2D model to test the hypothesis of zero wall concentration. Using a first-order reaction, we developed a criterion for external mass transfer control in terms of what was effectively a Damköhler number. In summary, it was quickly demonstrated that the experiments reported by Bennett et al.[16,36] were in the kinetically controlled regime, and thus that their analysis was faulty. Furthermore, in a key point, we demonstrated that if you take the experimental results (conversions) predicted by the full 2D model, and analyze them using the zero wall concentration assumptions, you will indeed predict low values of Sh that are consistent with the values that they reported. Ullah et al.[37] based their work on a kinetic expression for CO oxidation from Doory,[40] which was a Langmuir-Hinshelwood-Hougen-Watson (LHHW) style model of the classical Voltz et al.[41] form. Again, using that kinetic model in our 2D simulator showed that the wall concentrations should not be expected to be zero, and therefore the initial assumption was also invalid. So, at this point, we were able to say that the incorrect assumption of zero wall concentration was the culprit in the artificially low Sh (and corresponding Nu) values, and thus it was safe to conclude that values of the order predicted by classical heat transfer theory were indeed correct.

These two investigations essentially curve fit the apparent mass transfer coefficients. Using the kinetics that they used, a 2D model will over predict the conversion; thus, the kinetic models must not be correct. If one assumes that the washcoat diffusion resistance is zero (as they did), then it is necessary to reduce the apparent rate by external mass transfer to match the experimental conversions. Since that approach is not correct, it implies that washcoat diffusion is not negligible. We discuss washcoat diffusion in detail in a subsequent section, however, suffice to say here, that when the published kinetic data were reanalyzed allowing for washcoat diffusion, it was possible to match the experiments with the model.

In conclusion, this paper,[17] published in *Chemical Engineering Science*, can be considered a landmark paper because it really settled the controversy over the mass and heat transfer from a broad perspective. Although it may appear obvious, a posteriori, that the mass transfer limited assumptions were the cause of the error, what is obvious is not always known, and this paper finally squared the circle and brought consistency to this issue.

After about 1994, references to Hawthorn and Votruba models are few, and they are not used in serious work. Having said that, they may still be seen, and it is important to emphasize strongly that they not be used in any circumstances, because they are wrong.

New sets of models emerged from about 1995, which were correctly based on local rather than average values for Nu and Sh. These correlations have the following generic form:

$$\mathrm{Nu} = A\left[1 + B(\mathrm{Gz})^n exp\left(-\frac{C}{\mathrm{Gz}}\right)\right] \tag{9}$$

where the constant A is the asymptotic value for fully developed flow, and B, C, and n depend on the shape and wall boundary condition. Values for circular[3,42,43] and - non-circular channels[44,45] are summarized in Hayes et al.[24] for boundary conditions of constant wall temperature and flux.

The classical solutions for the Graetz problem use either constant temperature or constant flux for the wall boundary conditions. In truth, the chemical reaction at the wall gives a boundary condition between the two, and it has been proposed[44] that the real value can be found using the following interpolation formula[46]:

$$\frac{\mathrm{Nu} - \mathrm{Nu}_H}{\mathrm{Nu}_T - \mathrm{Nu}_H} = \frac{\mathrm{DaNu}}{(\mathrm{Da} + \mathrm{Nu})\mathrm{Nu}_T} \tag{10}$$

where Da is the Damköhler number, which for a first order reaction is given by the following:

$$\mathrm{Da} = \frac{\eta k_S D}{4D_a} \tag{11}$$

This formula implies that the limiting value for Nusselt number under kinetic control (low value of Da) is Nu_H, whilst as Da approaches infinity (complete mass transfer control) the Nusselt number approaches Nu_T. Using local values for Nu and Sh can give reasonable agreement between 1D and 2D models,[12,13,43] provided that the values obtained from the 2D model are used for the 1D model.

When the reaction rate is low, the system behaves like one with constant wall flux, and for high reaction rates it is akin to constant wall temperature. Thus, when a reacting stream experiences an ignition, at that point there can be a relatively sudden change in the value of the Nu. Early papers[11] showed a spike in the Nu versus Gz curve at this point, although we never observed it in any of our simulations.[15,17,47] The Houston group has done much theoretical analysis of Nu and Sh and has reported that this jump is in fact a numerical artefact.[48–50]

The Graetz problem assumed constant physical properties, as does much other reported work. Using our 2D model with the Navier–Stokes equation and temperature dependent physical properties, it became clear that the classical Nu versus Gz^{-1} plots were not unique, but rather a function of velocity, reaction rate, and other variables.[50] This variability was a result of the temperature dependent physical properties. More[22] performed a detailed study of Nu and Sh in monoliths, which included variable properties, which was continued by others afterwards.[24,25] All of this work confirmed that the entry length Nu and Sh are a function of the properties, and thus the sub-models represented by Equation (9) are not generally unique. The latest work[25] also showed that it is important to include the inlet region modelled as a contraction to capture the entry length effects correctly, and proposed correlations for Nu in the form of Equation (9) where the values of B, C, and n could be represented by functions.

Another very interesting observation for the case of variable physical properties with constant wall temperature was a minimum value of Nu smaller than the asymptotic value in the entry region.[22,25,50] The size of the minimum depended on the operating conditions, and an equation was proposed to account for this effect.

Most of the models developed are based on laminar flow. In reality, a certain degree of turbulence may be there, which will enhance the heat and mass transfer rates. The flow is, for most applications, naturally turbulent when entering the monolith and a turbulent to laminar transition takes place over some distance in the channel. Turbulence can also be generated by the monolith walls at the entrance and also by the surface roughness of the channel walls. Not much work has been done on turbulence in monolithic reactors, since the techniques available to turbulence intensities are difficult to use in small channels. The effects of decaying turbulence in the entry region of the monolith of the value of Nu has been shown.[51]

To summarize the entry length problem, it is still not actually obvious what is the best form of correlation to use in this region, and it is certainly much more complicated than originally assumed. It might not be extremely critical to capture this region correctly, except when the entry region accounts for a significant amount of the reaction or the monolith is very short.[52] The latter case is more important in laboratory scale reactors that are often used for collecting data used for kinetic modelling. In some cases,[53] we used a 2D model for evaluating kinetics where a correct representation of the entrance length was required. Most of our kinetic studies with monolith reactors, however, used a 1D model.[54–58] In kinetic optimization studies, a simple model is necessary to reduce computational cost. In kinetic studies, the aim is to operate in the absence of heat and mass transfer limitations, and therefore a simple model is justified.

As a final note on the selection of Nu and Sh, recall that for non-circular cross-sectional shapes, the Nu and Sh vary around the perimeter of the channel.[29] This effect is rarely, if ever, considered in 1D models. Using a pseudo-3D model with fully developed flow, however, we have shown the degree of variation for some typical realistic washcoated channel shapes with a chemical reaction, and even for a circular shape with a non-uniform washcoat distribution.[59]

3.5 | Modelling the washcoat diffusion

The washcoat is similar to any porous supported metal catalyst. The main difference with a typical packed bed pellet is that the maximum diffusion path is much smaller. The typical washcoat thickness varies from 10–150 μm[17] compared to pellet sizes of 6-mm diameter and more. Standard analysis of diffusion and reaction in porous catalysts models the catalyst as a continuous porous medium with an effective diffusivity. Actual reaction rates depend on the value of the effectiveness factor, which in turn depends on the Thiele modulus.[60] Much of the early work assumed that the internal diffusion resistance was negligible because of the thinness of the washcoat.[14,35,61,62] Two interesting exceptions are worth noting. One study used a simplified reaction scheme to model automotive exhaust gas in a washcoat treated as a flat plate and showed strong internal diffusion limitation was present.[63] Chou and Stewart[64] modelled a non-uniform washcoat in a square channel as flat plates but with rounded corners.

Our 1994 paper[17] was the first effort reported to model diffusion in a real fillet-shaped washcoat, and, as discussed earlier in this paper, we showed that washcoat diffusion is quite important. This paper led to a detailed study of washcoat diffusion in real washcoat shapes[65] and included washcoat diffusion in a full single channel model for first-order, power law, and LHHW kinetic models. One significant result of this work was the attempt to develop simple sub-models for the effectiveness factor. There are classical analytical solutions for isothermal first-order reactions in flat plates, infinite cylinders, and spheres. The generalized Thiele modulus solution takes the characteristic length as the ratio of catalyst volume to the surface area, and for this definition the effectiveness factor plots look similar, with some

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disagreement for medium-sized Thiele modulus.[60] This approach was not seen to give acceptable results for the washcoat fillet[65] and different methods were proposed as a solution. The best option, which was valid for any power law model (normal kinetics) was an equation that used matching between the low and high Thiele modulus asymptotes[23] which gave equations of the following form:

$$\eta = \left[1 + \left(\frac{a}{\Phi^b}\right)^{-n}\right]^{\frac{-1}{n}} \tag{12}$$

The constants a, b, and n depend on the reaction order and fillet shape. Generally, we have found that a and b are both very close to 1, and thus a good general expression for normal kinetics is as follows:

$$\eta = \left[\frac{1}{1 + \Phi^n}\right]^{\frac{1}{n}} \tag{13}$$

The value of n can be determined by curve fitting data obtained from the 2D washcoat model. We have also determined that for realistic washcoat shapes the washcoat can be assumed to be isothermal, even for very exothermic combustion reactions. We also used the latter equation in a study of washcoated gasoline particulate filters with a non-uniform washcoat loading.[66]

Another interesting point of diffusion resistance is that one can place a layer of blank washcoat over the active one, and this can be used to control axial temperature gradients in some cases. This idea was apparently first proposed in 1992[67] and was tested by us both experimentally[53] and numerically.[53,65] Diffusion barriers can be very useful for controlling super-adiabatic temperature rises when the Lewis number is significantly different from one.

A general equation for the effectiveness factor given by Equation (13) is useful for single channel models, either 1D or 2D, and can easily be implemented as a simple sub-model. Whilst many 2D washcoat simulations may need to be run to find the value of n, once determined the SCM simulation is relatively rapid, and the 1D generalized Thiele approach is not required. This was very useful for our catalytic combustion work, where a single reaction was often used, but is less useful for the case of multiple reactions or for LHHW type kinetic models. In this case it is necessary to solve the effectiveness factors contemporaneously with the SCM simulation. If it is desired to take the washcoat non-uniformity into account, then another method must be found.

In an early paper,[68] Aris proposed a method for calculating an average effectiveness factor for a bed packed with catalyst particles of different sizes, using a weighted size average. Papadias et al.[69,70] adapted the method for a non-uniform washcoat. They simply divided the washcoat into a set of slices of different thicknesses. The 1D diffusion reaction equation was solved in each slice as a 1D problem, and then the average was computed. The idea is quite clever, because solving a set of 1D problems is computationally much cheaper than solving a 2D problem. However, when we applied the method and compared it to a 2D solution we observed some discrepancy between the two results, especially for very non-uniform washcoats. This is likely because of the diffusion between sections. To improve the method, we discovered that, rather than using the real geometrical length of each section, we could compute an effective or apparent length. This was easily computed from a single 2D simulation using a first-order reaction, by back-calculating the apparent length from the corresponding effectiveness factor for the section using the 1D analytical solution. Using this apparent length rather than the geometric length gives marked improvement in the results.[71] It is interesting that the effective length computed from the first-order reaction worked equally well for LHHW style kinetics.

LHHW style kinetic models have the possibility of effectiveness factors larger than 1. This artefact can also give numerical problems in the solution and can be time consuming. We have therefore developed fast approximation methods for this type of kinetic model.[72]

It is now widely recognized that washcoat diffusion is significant in catalytic converters for many applications. The choice of model to use ranges from the full 3D model, incorporated into a 3D SCM, or to use a simpler sub-model, which can range from 2D solution, an approximation of a 2D solution, or a 1D model based on an average thickness. The final choice depends on the accuracy of the solution desired, and the accuracy to which the necessary parameters are known. We have shown that taking the real channel geometry into consideration can influence the predicted light-off curve significantly under some conditions.[73]

As a final point, it is obviously essential to have good values for the effective diffusion coefficient. Typically, this must be done experimentally, but to make the measurements on real washcoats is quite tricky. We developed two techniques to do this. The first was based on a method proposed by Beeckman[74] that involves measuring diffusion across washcoated monolith channels, where the gas in alternate channels flows counter-currently.[75] The other method also measured diffusion across real monoliths using a modified form of the classical Wicke-Kallenbach cell.[76] These methods can be time consuming, because it is necessary to measure the effective diffusion coefficients in both the substrate as well as the washcoated monolith. However, reliable values are necessary to model the diffusion effect accurately.

4 | THE FULL-SCALE MONOLITH MODEL: ECM

4.1 | Description and challenges

The SCM can now be modelled at a reasonably high level of complexity, although solving the complete 3D model may still be too computationally costly for some situations. If all of the channels in the full monolith reactor had identical conditions, then a SCM could be considered to be representative of any channel, and hence represent the performance of the full-scale system. This would be nice, because we could then model the entire converter at a good level of detail. Unfortunately, it is not the case. Most monolith reactors form part of a complex system. The inlet and outlet pipework are usually of smaller size than the monolith, so diffusors are used at each end of the monolith. These are often conical shaped, which yields flow recirculation and causes significant flow maldistribution. Although monolith reactors are usually insulated, there is inevitably heat loss and the resulting radial variations in temperatures. Thus, modelling the entire converter as a single channel is not realistic,[77] and the entire reactor must be modelled for an understanding of the real system, using the so-called ECM. There has been significant effort in the literature to develop models for a complete catalytic converter including the inlet and outlet diffusors. A good review of the early history of ECM models, and the different types of models studied can be found in our first three papers in which we presented an ECM model,[78–80] and also in one of the earliest works on ECM published in 1989.[81] We offer a very brief overview here of the main choices that need to be made.

As noted earlier in the paper, a realistically sized catalytic converter contains thousands of channels, each with its catalytic washcoat. A full spatial discretization of the channels, washcoat, and walls is still prohibitively expensive. Just to illustrate the scale of the problem, as an academic exercise, we modelled the flow through a full-scale converter of 7539 channels, with every wall and channel discretized. This problem is too much for a typical finite volume or finite element code, so we used the Lattice Boltzmann method with 12 billion lattices. Running on 512 cores, the resolution of the flow field took about three weeks of CPU time, in addition to about a week for pre-processing.[82] Not to mention that the flow was laminar, and without reactions or heat transfer. So, whilst this was a fun exercise, it is clearly not a practicable one.

The main model reduction technique used is to represent the monolith as a continuous porous medium. With this representation, all spatial granularity is lost, so there is no differentiation of fluid and the two solid phases (substrate and washcoat). The use of a continuous porous medium does present a few challenges. One is whether or not to use a homogeneous or a heterogeneous model. The former does not differentiate between fluid and solid phases, whilst the latter solves the conservation equations for each phase, and couples them using heat and mass transfer coefficients. Our first paper using an ECM[78] investigated the use of reverse flow reactor to destroy fugitive methane emission for heavy duty trucks with lean burn natural gas engines. It was a collaborative effort with Fluent in which we developed a dual zone heterogeneous model. During this work it was evident that a heterogeneous model was necessary, especially for rapid transients.

Accepting that a heterogeneous continuum model is necessary for computational efficiency, we can then identify perhaps three significant challenges in the development of a reliable model. These can be listed as requirements for the correct specification of (1) momentum balance equation, (2) effective thermal conductivities, and (3) methods to include the diffusion and reaction in the washcoat. We now consider these points in turn.

4.2 | Solving the momentum balance equations—Velocity profiles and pressure drop

One of the main purposes of using an ECM is to capture the velocity profile correctly, which is in turn related to an accurate representation of the pressure drop. For flow in a porous medium, the parameter that governs pressure drop is the permeability. For laminar flow, the relationship between velocity and pressure drop is linear, whilst for turbulent flow it is quadratic. The naïve approach to setting the permeability is simply to assume fully developed laminar flow in the monolith, and then back-calculate the axial permeability using the Hagan-Poiseuille equation. To ensure unidirectional flow, set the radial permeability two or three orders of magnitude lower than the axial one.[78] This was basically the approach for our second paper with a continuum model,[79] which was meant to settle a few outstanding questions and included validation of the velocity profiles

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using literature data.[83] The agreement was reasonable, although not perfect. We mention here one interesting result: it has been suggested in the literature that advantageous operation can be achieved by using a composite monolith composed of a radially variant cell density.[84] The idea is that placing annular rings of decreasing cell density around a high cell density core would lead to a more uniform flow distribution and hence better light-off. Although we did show better flow distribution, the light-off behaviour was not as good as the monolith with uniform cell density. Considering that Toyota now markets a converter of this design, there is possibly more work to be done on these simulations.

Along with the questions about the permeability model, there are also various choices to make as to the form of the governing momentum balance equations in the porous medium. As a result of a study of a novel converter design,[85] we realized the importance of getting this right, and therefore conducted an extensive study aimed at the developing the correct model for the momentum balance in the converter, which considered all of the essential aspects of the problem and would yield a good value for the pressure drop and the flow distribution. This work was published in eight papers.[86–93] Two factors are of key importance: (1) the correct representation of the flow regime in the porous medium and (2) the correct permeability model, which is related to the pressure drop.

According to the fundamentals, inside of a monolith channel, the pressure drop comes from the shear stress between the flow lamina moving at different relative velocities. That is, the phenomenon depends on the channel velocity profile. Variables affecting such profile are the inlet velocity profile, hydraulic development of the flow, and turbulence. Some others can come from the flow colliding with the frontal face of the substrate and when it leaves the monolith. Many of those effects are considered minor losses in the classical literature, however, because of the remarkably low pressure drop produced by a monolith; they may not be negligible in, for example, short monoliths.[90,94]

Consider the flow regime inside of the monolith. Assume that the monolith forms part of a system, including upstream and downstream diffusors and pipes. The flow in the latter devices is normally turbulent, but within the porous medium it is expected for the most part to be laminar. Typically, the problem is solved using Reynolds averaged Navier–Stokes (RANS) (e.g., k- ω) models, which, to preserve continuity of mass and momentum, is applied to both the empty and the porous zones. This approach allows for turbulence in the porous zone, that is, the turbulence viscosity will be different from zero in the porous laminar zone. To eliminate this, commercial software typically sets the turbulence viscosity to zero in this porous zone, which eliminates the turbulence generation; however, the existing turbulence is simply transported from the entrance to the exit, which is not physical. Alternatively, one could artificially set the values of k and ω to zero, but then the turbulence will remain zero in the downstream section. Therefore, we developed a method to model the flow regime correctly.

The first step was to investigate the behaviour of a highly turbulent flow entering monolith channels with sub-critical Reynolds number. The main concern was that the approach of completely neglecting the turbulence once the flow enters the substrate somehow violates momentum conservation. A more physical approach was proposed, as a result of the study of the changes on the turbulence vortices and kinetic energy. We noticed from data obtained using large eddy simulations (LES) simulations at the channel scale that turbulence dissipates quickly inside of the channels, provided that there is low channel Reynolds.[86,87] This allowed the development of sink terms for the motion equations, that give the porous medium a realistic physical behaviour of the turbulence. The study of the changes on the flow regime because of the presence of a monolith was finalized by investigating flow leaving a series of channels of different shapes and sizes. It was found that after passing the rear face of the substrate, two phenomena, the flow around the last part of the substrate and the flow acting as a jet, can generate turbulence downstream of the monolith.[88] Initially, it was observed that if the flow is steady, then a relatively high velocity is required to trigger turbulence in that zone. Later, the investigation was expanded to consider pulsating flow inside of the channels.[90] In that situation, it is much easier to observe turbulence generation downstream of the substrate. That, of course, can be especially relevant when several monoliths are placed in series. Both dissipation and generation of turbulence due to the monolith were put together in a new approach to model the flow regime through a substrate that correctly accounted for the turbulence.[89]

Having unveiled the behaviour of the flow regime through a monolith, the only missing part was to assess its impact on pressure drop. In that sense, posterior studies about turbulent flow approaching a monolith led to striking results. We observed that, after the laminarization of the flow, it does not remain steady, but it turns into a pulsating regime that persists along the entire substrate.[92] That research also concluded that the backpressure under a pulsating regime was almost the same as that obtained from steady flow; hence, no further corrections are necessary to account for the extra pressure drop due to the upstream turbulence.

The pressure drop for flow through monoliths modelled as a porous medium is computed by setting an apparent permeability that produces the same pressure drop as when the flow passes through a channel. That is, a source term based on Darcy's law is added to the momentum balance. The exact mathematical expression is as follows:

$$S_{u_i} = -\frac{\mu}{\alpha_i} u_i \tag{14}$$

where the parameter α_i is the apparent permeability. It is different in the radial and axial directions. For fully developed laminar flow, α_i is constant and a function of the friction factor, as $\alpha_i = \phi f_D / C$. However, when accounting for the entrance length, the apparent permeability changes with the axial coordinate. Prior to this work, there were models for friction factor in developing flow, but for circular pipes starting from a flat velocity profile. The issue was that monolith channels are not always circular and do not start from a flat profile. So, we carried out a series of computational experiments of laminar flow entering straight into monolith channels with several cross-section shapes, wall thicknesses, and cell densities. From the result, we observed that the inlet velocity profile can be either concave or convex, which affects the formation of the boundary layer at the beginning of the channels.[90,93] The entrance friction factor differed significantly from that of a flat profile. Fortunately, it was only slightly sensitive to the monolith void fraction for open frontal areas between 0.4–0.8, so it was not necessary to include it as a variable in the final model. Another important point was that the diameter of the *vena contracta* in the cases analyzed was practically the same as that of the channel. The data obtained were used to develop a new expression for friction factor for monolith channels with developing flow, as follows[93]:

$$f_{(x)} = \frac{1}{\mathrm{Re}}\left[\left(c_1\sqrt{\mathrm{Gz}}\right)^n + \left(f_D \mathrm{Re}\right)^n\right]^{\frac{1}{n}} \tag{15}$$

The initial idea was to have values for c_1, n, and f_DRe for every channel shape. However, after analyzing several channel shapes, we noticed that when scaled by their respective asymptotic value, the curves for all the shapes investigated collapse to a similar one (see Figure 3).

The friction factor can ultimately be used to obtain an expression for the apparent permeability of the equivalent porous medium[93]:

$$\alpha_{(x)} = \frac{\alpha_D f_D}{f_{(x)}} \tag{16}$$

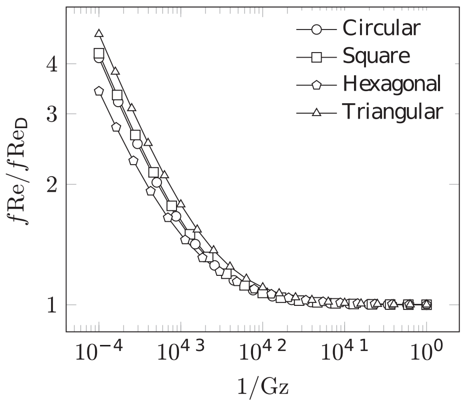

FIGURE 3 Scaled *f*Re curves for several channel shapes. Reprinted with permission from Elsevier[93]

The last two phenomena analyzed were the losses at the frontal and rear faces of the substrate. Let us call them the inlet and outlet effect, respectively. At the inlet, the losses come from collision and friction, consequently observed to be proportional to both velocity and square velocity. We noticed that this pressure drop was a function of the open frontal face of the substrate, changed depending on the channel shape, and took place in a thin area prior to the entrance of the substrate. Based on that, we proposed to model it as a porous jump. That is, a sudden decrease of mechanical energy, with a magnitude computed as follows:

$$\Delta p_{in} = f_1|_{in} \mu u_i + f_2|_{in} \frac{1}{2} \rho u_i^2 \tag{17}$$

For the flow leaving the monolith, the losses can take place during a relatively long distance downstream of the substrate; hence, their modelling is much more challenging. For simplicity, and given that they are explained by friction only, we proposed the following expression to estimate the outlet effect on pressure:

$$\Delta p_{out} = f_{2out} \frac{1}{2} \rho u_i^2 \tag{18}$$

The combination of equations describing the variable permeability and the inlet and outlet effects was then presented as a new approach to model the pressure drop through a monolith reactor. The model parameters and additional details are summarized in two papers.[90,93] Excellent agreement was obtained with experimental data.

We have applied a similar methodology to analyze a wall flow filter. Wall-flow filters are comprised of monoliths with porous walls, but with alternate ends of the channels plugged at their inlet or outlet, forming a chessboard pattern. With that configuration, the flow that enters the filter by one channel leaves it by a different one, passing through the porous walls of the filter in the process. That imposes some extra challenges when modelling the total pressure drop. First, the backpressure from the flow crossing the filter walls must be accounted for as an additional effect. Second, as the gas advances through the filter, the flow rate in the inlet channels decreases progressively, at the same time that it increases in the outlet channels. That is, developed flow cannot be stated based on the analysis of the velocity profile only. In industry, the interest is in a model of the entire filter; however, similarly to the unplugged monoliths, most of the aforementioned phenomena must be studied in a channel scale. We performed a series of CFD studies of flow through this type of particulate filter, considering a group of four channels of different shapes and wall permeability. Based on the results, we proposed the use of a dimensionless velocity profile to state fully developed flow, new models for friction factor, inlet and outlet losses, and momentum correction, together with other analyses of the flow inside of the filter.[95] This work was experimentally validated.

To conclude this section, what we have achieved for the first time over the past 3 or 4 years is a completely consistent flow and pressure drop model for the simulation of a monolith reactor as a continuous porous medium.

4.3 | Modelling the effective thermal conductivity of the monolith

As noted in the introduction, temperature variations in the monolith reactor will be pronounced. The primary mechanisms for heat transfer in the monolith brick are conduction in the radial (transverse to flow) direction and convection and conduction in the axial (aligned with the flow) direction. In the first instance, let us consider the apparent thermal conductivity of a porous medium. There is a vast amount of literature on the modelling of apparent thermal conductivities, and the value depends on the nature of the medium, including its structure. It is not obvious a priori what is the best model for a given structure. The choice will depend on whether a homogeneous or heterogeneous model for the porous medium is used, with the homogeneous model somewhat easier to conceptualize. Most reported work is based on a homogeneous model. Remember that in the continuum model, all spatial discretization is lost.

Consider first a homogeneous model for the porous medium. The energy balance in terms of effective thermal conductivities is as follows:

$$\frac{\partial}{\partial x}\left(k_{r,\mathrm{eff}}\frac{\partial T}{\partial x}\right)+\frac{\partial}{\partial y}\left(k_{r,\mathrm{eff}}\frac{\partial T}{\partial y}\right)+\frac{\partial}{\partial z}\left(k_{z,\mathrm{eff}}\frac{\partial T}{\partial z}\right) - \nu_s\rho_f C_{P,f}\frac{\partial T}{\partial z}=0 \tag{19}$$

The x and y coordinates denote the radial position, and the z coordinate the flow direction. For an isotropic porous medium, the effective thermal conductivity does not depend on direction, and a single value is used. The most common model for the effective thermal conductivity, k_{eff}, is a simple weighted average of the thermal conductivities of the solid, k_S, and fluid, k_f, phases:

$$k_{\mathrm{eff}}=\phi k_f+(1-\phi)k_S \tag{20}$$

where φ is the porosity (void fraction). This model, which is the default model in Fluent, is only valid for an isotropic porous medium with a single solid phase, neither of which is the case for a monolith, for which we must use different values for the two directions. Development of thermal conductivity models specifically for washcoated monoliths has been addressed in four interesting papers.[96–99] These papers represent the blank and washcoated channels in an idealized form, and then approximate the effective conductivity using the time-honoured concept of series and parallel resistances,[27] with different combinations being tested. Resulting models depend on the solid and fluid thermal conductivities, as well as the volume fractions of each material. Comparisons of the approximate models with the full numerical solutions for the real domain, both for a single row of cells as well as webs of cells of different shapes,[98] showed that the agreement depends on the resistance model used, as well as the physical characteristics of the monolith, but in all cases the agreement is generally acceptable. The interested reader should refer to the cited literature for the equations for the radial thermal conductivities, but we mention here that in the axial flow direction, the effective thermal conductivity can be represented by the classical model for the blank monolith:

$$k_{z,\mathrm{eff}}=\phi k_f+(1-\phi)k_S=\phi k_f+\lambda k_S \tag{21}$$

where λ is the volume fraction of substrate. For a monolith with a volume fraction of washcoat equal to ξ, the equation is as follows:

$$k_{z,\mathrm{eff}} = \phi k_f + k_S\lambda + k_W\xi \tag{22}$$

These models are good for a homogeneous model; however, the models for the heterogeneous models are more problematic. In this case, the two energy balance equations are as follows:

$$\frac{d}{dz}\left(k_{zf,\mathrm{eff}}\frac{dT_f}{dz}\right) - \nu_s\rho_f C_{Pf}\frac{dT_f}{dz} + ha_v\left(T_S - T_f\right) = 0 \tag{23}$$

$$\frac{\partial}{\partial x}\left(k_{rs,\mathrm{eff}}\frac{\partial T_s}{\partial x}\right) + \frac{\partial}{\partial y}\left(k_{rs,\mathrm{eff}}\frac{\partial T_s}{\partial y}\right) + \frac{\partial}{\partial z}\left(k_{zs,\mathrm{eff}}\frac{\partial T_s}{\partial z}\right) + ha_v\left(T_f - T_S\right) = 0 \tag{24}$$

Now we require effective thermal conductivities for each phase. In the axial direction, it is evident that we can define the following (based on the resistance in parallel concept):

$$k_{zf,\mathrm{eff}} = \phi k_f \tag{25}$$

$$k_{zs,\mathrm{eff}} = k_S\lambda + k_W\xi \tag{26}$$

The correct value in the radial direction is less evident, and further work should be done in this area. The best option may be to use the effective radial thermal conductivity for the homogeneous model in the solid phase energy balance equation, and to set the radial thermal conductivity in the fluid phase balance equation to zero or leave it at the value for the gas. We should note that these correlations have also been used in other monolith applications, for example, with high thermal conductivity substrates.[100]

4.4 | Incorporating the inter-phase and intra-phase heat and mass transfer

The two modes of heat and mass transfer and their importance have been extensively discussed. The use of the continuum model means that the conservation equations for species and energy are essentially 1D in nature, although not exactly the same as the 1D SCM discussed previously. The fluid phase temperature and concentrations are volume-averaged quantities, although depending on the discretization they are not exactly equivalent to single channel mixing cup averages. Notwithstanding that observation, the fluid and solid temperatures and concentrations are coupled using the same Nu and Sh correlations discussed earlier in the paper. As noted earlier, for non-circular channels, the Nu and Sh may vary around the perimeter, but this is not implemented in the continuum model and a single average value is used for each axial location.

The inclusion of washcoat diffusion must be done via a sub-model of some type. Again, because all spatial granularity is lost, the non-uniformity of a washcoat cannot be included directly, and therefore, for any given point in the reactor, a call must be made to the sub-model to compute the value of the effectiveness factor or the global reaction rate. The methods used by the sub-model are effectively the same as those described previously and can range from a simple algebraic equation to the solution of a 2D diffusion reaction problem in a non-uniform washcoat. The latter sub-model can add very significantly to the cost of the simulation.

5 | ALTERNATIVE METHODS FOR MODEL REDUCTION

We have discussed using high fidelity models to find sub-models, often in the form of equations that use a few parameters to describe complex phenomena (e.g., Nu, Sh, or the effectiveness factor). A similar idea exists in the development of rate equations, where assumed mechanisms are reduced to fairly simple rate expressions expressed in terms of gas phase species. The LHHW methodologies are examples. However, there are cases, both in the context of reaction rates and heat and mass transfer, where simple equations cannot be developed, or their use is not desirable. In this case, other model reduction strategies can be used; they are briefly discussed here.

One strategy that can be used to reduce the complexity of the multi-scale problem is to use pre-computed solutions for the sub-models and, for example, store the values in a look-up table or use them to train a neural network.[101–106] The cost of building the look-up table depends on the complexity of the underlying model, but once built it is very cheap to access. Pre-computed rate data for complex reacting flows have been used in single channel monolith systems for complex kinetics.[104] That work was the foundation for investigating cases of complex washcoat design and reactor analysis.[105–107] Taking the model reduction to the next level, Nien et al.[108] developed a detailed methodology for the systematic multi-scale model reduction for methane oxidation in a full-scale converter. Successive look-up tables have included detailed mechanism-based rate models, washcoat diffusion in a fillet-shaped washcoat, and external heat and mass transfer. Speed-up factors of many thousands were observed. A similar methodology allowed for our investigation of the effect of changing monolith cell density on reactor performance whilst keeping all internal factors consistent.[80]

The main drawback to look-up tables is that they require increasingly large amounts of storage space as the number of variables increases. The maximum practical limit is about seven, depending on the interpolation methods. An alternative is the use of a neural network. We have experimented with this option and have seen good potential, but nothing has been published. This is an area of promising future research.

6 | CONCLUDING REMARKS

As we have seen, in 1990 the level of understanding of the following key points was either non-existent, minimal, or subject to controversy:

- the role of washcoat diffusion;
- the correct values of Nusselt and Sherwood number for use in SCM;
- the correct way to model the momentum balance in a continuum ECM;
- the best values for the effective thermal conductivities in a continuum ECM;
- the importance of solving the momentum balance equation in the SCM; and
- the correct inlet and outlet boundary conditions for the SCM, especially as they relate to the momentum balance equation.

These points comprise most of the critical factors in the modelling of monolith reactors. Looking back over 30 years, it is easy to see the great strides that have been made in the modelling of monolith reactors. That is not to say we now have a complete understanding of each of these factors, or that we can stop improving our models. To quote George Box: 'Essentially all models are wrong, but some are useful'.[109] So, all that we will commit to now is that the models that we have today are less wrong and more useful.

It is also fair to say that many of the assumptions used in the early days of this period were made to overcome limitations of computational resources. As these limitations eased, models have included more and more features, and will likely continue to do so over the next few years.

It is also interesting to reflect on the acceptance of modelling by industrial users, as opposed to academics. Advanced computational modelling has been widely used since WWII, but was slower to take hold in the chemical related industries. Certainly, during the 1990s, there was scepticism by the real end user, possibly because of the lack of sophistication and robustness of the models then extant. That situation has certainly changed, especially in the automotive emissions sector. One of the most powerful signs of this was seen in the launch in 2009 of the workshop series on MODelling of Exhaust Gas After Treatment (MODEGAT), now held every two years in Bad Herrenalb, in the heart of the Black Forest. Attracting 100–120 participants each time, with about half from industry, they showcase the latest advances in modelling. Its counterpart in the United States is CLEERS (Crosscut Lean Exhaust Emissions Reduction Simulations). That initiative is funded by the U.S. Department of Energy (DOE) to promote the use of computational tools in EGATS. It is also interesting to observe the increase in the modelling presentations at the CAPoC (Catalysis and Automotive Pollution Control) conferences in Brussels.

So, what does the future hold? Not having the powers of Nostradamus, we can only guess. It is likely that the internal combustion engine will be around for many years to come, so the need for EGATS will continue. Monolith reactors (and other structured types) will likely continue to find other uses, and their presence will expand. Computers and software will continue to increase in power. Therefore, it is probably safe to assume that there will continue to be advances in this area for many years to come, and there will many more gatherings at the Haus der Kirche.

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As mentioned in the paper, the early reactor simulation codes were finite element programs written in Fortran. These codes had their genesis in the thermofluid FEM codes developed by Philippe Tanguy (POLY2D) and later Francois Bertrand and others in that group. REH also collaborated extensively with Francois on aspects of monolith modelling over the years.

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NOMENCLATURE

a	constant
a_v	surface area per unit volume (m^{-1})
A	constant
B	constant
b	constant
C	constant
c	constant
c_1	constant
C_P	heat capacity ($J\ kg^{-1}\ K^{-1}$)
D	diameter (m)
D_a	molecular diffusion coefficient ($m^2\ s^{-1}$)
Da	Damköhler number
f	friction factor
f_1, f_2	model sub-functions
h	heat transfer coefficient ($W\ m^{-2}\ K^{-1}$)
k	thermal conductivity ($W\ m^{-1}\ K^{-1}$)
k_S	first order rate constant ($m\ s^{-1}$)
L_{ent}	entrance length (m)
L	length (m)
n	constant
Gz	Graetz number
Nu	Nusselt number
Nu_H	Nusselt number with constant wall flux
Nu_T	Nusselt number with constant wall temperature
Pr	Prandtl number
p	pressure (Pa)
Re	Reynolds number
S_{ui}	momentum source term ($m\ s^{-2}$)
Sc	Schmidt number
Sh	Sherwood number
T	temperature (K)
u	velocity ($m\ s^{-1}$)
u_i	velocity vector ($m\ s^{-1}$)
ν_s	superficial velocity ($m\ s^{-1}$)
x	coordinate (m)
y	coordinate (m)
z	coordinate (m)

Subscripts

f	fluid
r	radial
S	solid
eff	effective
W	washcoat
z	axial

Greek symbols

α_i	apparent permeability (m^2)
η	effectiveness factor
Φ	Thiele modulus
ϕ	porosity
ρ	density ($kg\ m^{-3}$)
μ	viscosity (Pa s)
λ	volume fraction of substrate
ξ	volume fraction of washcoat

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